

- The following rules apply:

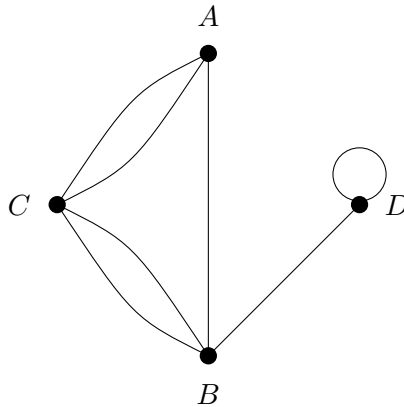
- If a question is followed by work, then work of some form is required. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. A correct answer that has no supporting work when work is required will not receive full credit.
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- If a question has multiple parts, the points for that question are divided evenly among the parts.
- Do not write in the ✓ + − box, because this is for grading. The grading scale (except True/False) is: ✓ = 100%, + = 50%, − = 0%

[illegible]

1. (90 points) **True/False** Circle T to indicate the statement is always true, or F otherwise.
Note: homework protection does not apply to True/False questions.

- T F** (a) K_9 has an odd number of even vertices.
- T F** (b) A complete graph with 4 vertices has 6 edges.
- T F** (d) A regular graph that has 10 vertices and 40 edges has an Euler Circuit, unless it is disconnected.
- T F** (e) $ABCD A$ and $CDABC$ describe the same tour.
- T F** (f) If G is a complete graph with n vertices, then G has $\frac{n(n-1)}{2}$ edges.
- T F** (g) The Nearest Neighbor Algorithm always gives the optimal tour of a graph with Hamilton circuits.
- T F** (h) The Brute Force Algorithm always gives the optimal tour of a graph with Hamilton circuits.
- T F** (i) A graph with a Hamilton circuit has a Hamilton path.
- T F** (j) There exist a graph with an odd number of odd vertices.

2. (42 points) Use the graph below to answer the following seven questions.

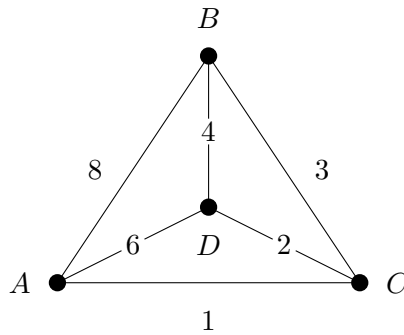


- (a) ☐ ☐ ☐ Write the vertex set: $V(G) =$
- (b) ☐ ☐ ☐ Write the edge set: $E(G) =$
- (c) ☐ ☐ ☐ Which edges, if any, are adjacent to DB ?
- (d) ☐ ☐ ☐ Which vertices, if any, are adjacent to D ?
- (e) ☐ ☐ ☐ What is the degree of each vertex?
- (f) ☐ ☐ ☐ Which edges, if any, are bridges?
- (g) ☐ ☐ ☐ If an Euler Circuit or Euler Path exists, list one and write if it is an EC or an EP. If neither is possible, explain why.

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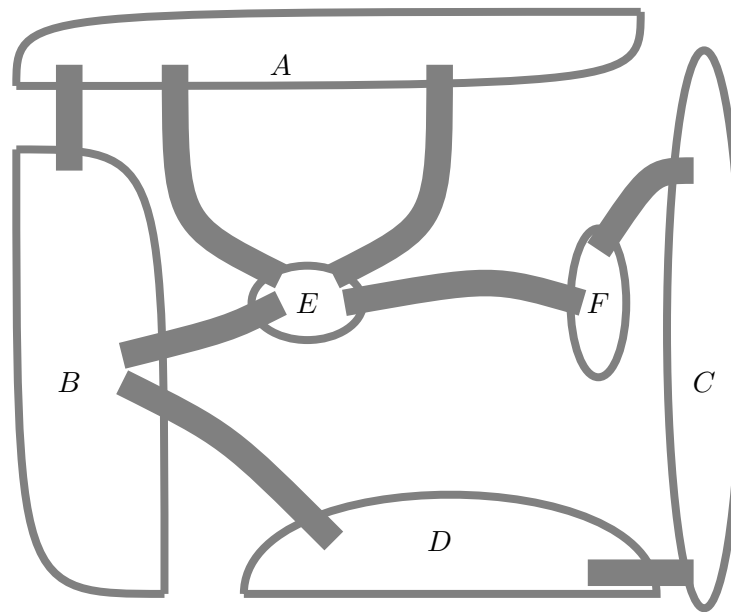
Show your answer by labeling the edges 1, 2, 3 and so on in the order they are traveled.

4. (60 points) Use the weighted graph below to demonstrate your prowess with the Repetitive Nearest Neighbor Algorithm.

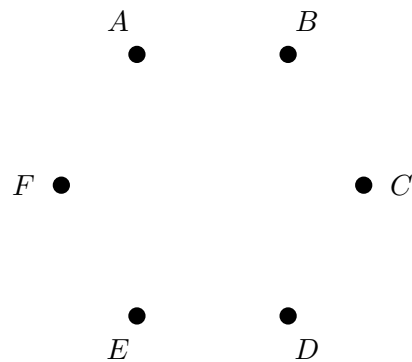


☐ ✓ ☐ + ☐ - What is the route and weight resulting from the application of the Repetitive Nearest Neighbor Algorithm?

5. (42 points) Use the map of islands and bridges shown below to answer the next three questions.



- (a) ☐ ☐ ☐ Draw a graph that models the islands (as vertices) and bridges (as edges).



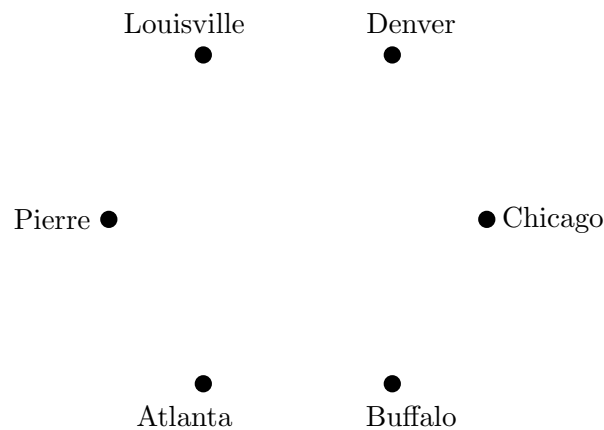
- (b) ☐ ☐ ☐ Is it possible to start a journey at D and end at B , crossing each bridge in the picture exactly once? Explain with a mathematical reason. (Note: a list of vertices is *not* a mathematical reason.)

- (c) ☐ ☐ ☐ If your answer in (b) was 'Yes', list the path. If your answer was 'No', which bridge(s) will be crossed twice for an optimal journey?

6. (60 points) The table below lists the mileage between 6 cities. Use the table to answer this question.

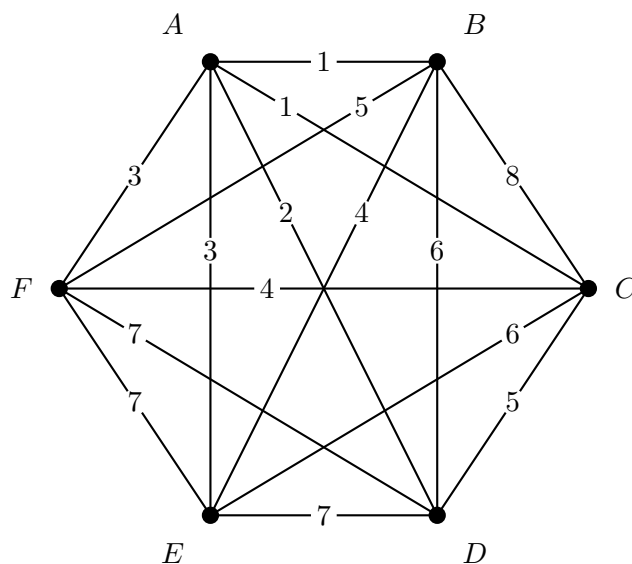
	Atlanta	Buffalo	Chicago	Denver	Louisville	Pierre
Atlanta	*	859	674	1398	382	1361
Buffalo	859	*	522	1508	532	1285
Chicago	674	522	*	996	292	763
Denver	1398	1508	996	*	1120	518
Louisville	382	532	292	1120	*	1055
Pierre	1361	1285	763	518	1055	*

☐ ☐ ☐ Suppose you are the head of a large company with offices in each city. Further, suppose that you want to send one of the managers at one of the offices in one of the cities to visit the offices in each other city (without visiting any city twice) and return home. You don't want to pay too much for gas, however, so you employ an algorithm to find an approximate solution. What is the result of the repetitive nearest neighbor Algorithm applied to this set of cities? (i.e. draw one optimal tour and state the total mileage)



7. (28 points) Occasionally, a situation may arise where the Nearest Neighbor Algorithm could produce multiple results. That is, it might be the case that there are multiple "cheapest" edges at a given vertex. The following weighted graph is an example of a situation where this might happen. Use the graph below to find two Hamilton circuits resulting from the Nearest Neighbor Algorithm (starting at F). For your convenience, the edge weights are also sorted.

Edge	AB	AC	AD	AE	AF	BE	CF	BF	CD	BD	CE	DE	DF	EF	BC
Weight	1	1	2	3	3	4	4	5	5	6	6	7	7	7	8



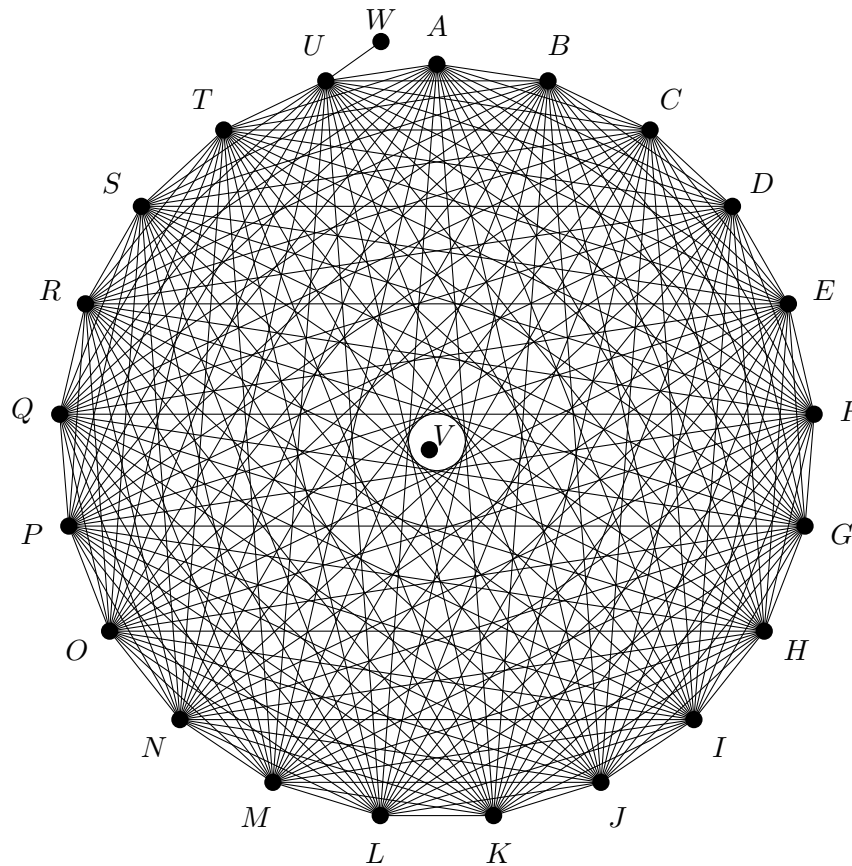
NNA tour 1	☒	☐	☐
A ● B ● F ● ● C E ● ● D			
NNA tour 2	☒	☐	☐
A ● B ● F ● ● C E ● ● D			

8. (42 points) Small drawing questions. You do not need to label the vertices. If you are directed to draw a graph that is impossible, say so and explain why. If the number of vertices is not specified, use no more than six; if the number of edges is not specified, use no more than eight.

- (a) ☒ + ☐ - Draw a graph that has a Hamilton Path but no Euler Path.
- (b) ☒ + ☐ - Draw a graph that has exactly four vertices and exactly two bridges.
- (c) ☒ + ☐ - Draw a graph that is not simple and not regular.
- (d) ☒ + ☐ - Draw a graph that is regular and connected, but has no Euler Paths, Euler Circuits, Hamilton Paths, or Hamilton Circuits.
- (e) ☒ + ☐ - Draw an example of a weighted graph with four vertices where the Brute Force Algorithm is quicker (fewer Hamilton Circuits are checked) than the Repetitive Nearest Neighbor Algorithm. (Hint: there is a *very easy* example of this)
- (f) ☒ + ☐ - Draw an example of a regular graph that has at least three vertices and an Euler Path.

9. (10 points) Use the graph below to answer the next five questions. If you aren't sure how many of something there are, at least give a range of possible values with an explanation.

(Note: It might be easy to miss, so be sure to notice the vertex V in the center of the graph)



- (a) ☐ $\checkmark$ ☐ $+$ ☐ $-$ How many bridges are in this graph?
- (b) ☐ $\checkmark$ ☐ $+$ ☐ $-$ How many Euler Paths are in this graph?
- (c) ☐ $\checkmark$ ☐ $+$ ☐ $-$ How many Euler Circuits are in this graph?
- (d) ☐ $\checkmark$ ☐ $+$ ☐ $-$ How many Hamilton Paths are in this graph?
- (e) ☐ $\checkmark$ ☐ $+$ ☐ $-$ How many Hamilton Circuits are in this graph?